Venkata Lakshmi Parimala Pasupuleti

Data Scientist · Machine Learning · Statistical Modeling · Python · SQL · AWS

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of production Python and SQL engineering, specializing in **ML model development, statistical modeling, and large-scale financial data pipelines**. Built and deployed ML models end-to-end — from feature engineering and training through evaluation, validation, and production deployment. Shipped Entrata's first production ML pipeline (PyTorch + OpenAI API) across **1.6M real financial records** with quality gates and compliance controls enforced before launch. Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries and **85%+ classification accuracy** on 104K+ records using Random Forest + Logistic Regression with cross-validated feature engineering. Applied Isolation Forest anomaly detection across 3 research pipelines, surfacing outliers manual review missed. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · ML Research · Anomaly Detection

- Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries by designing and training multi-source prediction models (Python, R) on WHO and UK Road Safety datasets — data collection, feature engineering, cross-validation, and pipeline automation end-to-end.
- Surfaced statistically significant anomalies **manual review missed** across 3 research pipelines by applying Isolation Forest — flagged outliers in public health and road safety datasets and translated findings into actionable research recommendations.
- Improved research comprehension **40%** across two cohorts by translating model outputs into decision-ready visualizations using Tableau, ggplot2, and Quarto — designed faculty leadership dashboards for ongoing data-driven program evaluation.

Entrata

Jan 2017 – Aug 2024

Software Engineer — Machine Learning & Data Science

- Designed, trained, evaluated, and deployed Entrata's **first production ML model** (Python, PyTorch) across 1.6M real financial records — feature engineering, output validation, and quality gates enforced before release; model productized into a revenue-generating customer feature.
- Modeled income eligibility for **millions of tenant certifications** across LIHTC/Section 42, Section 8/HCV, and HUD programs — AMI threshold classification, income-limit rule application, and household composition analysis; detected income misrepresentation and fraudulent documentation patterns across 100K+ unit portfolios with zero misclassifications.
- Built large-scale analytical data pipelines on **AWS** (S3, EC2, RDS, Glue) processing 1M+ financial records — optimized **200+ SQL queries** for 30% performance improvement, enabling faster model training and analysis cycles at 99.5% uptime.
- Translated complex financial domain logic into **automated decision models** (Python, PL/SQL) — income eligibility scoring, recertification workflows, and compliance threshold classification across 100K+ unit portfolios; zero violations over 7+ years of production operation.

PROJECTS

CrashClarity: Large-Scale Accident Severity Classification (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Cross-Validation — Live Dashboard

- Achieved **85%+ classification accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression ensemble — feature engineering across weather, road surface, light conditions, speed limits, and temporal variables; threshold calibration to minimize false positives.
- Surfaced top predictive features via feature importance analysis; deployed **interactive results dashboard** on GitHub Pages for non-technical stakeholders — translating model outputs into actionable severity-reduction insights.

Leasel: ML Evaluation Harness with Pass@k Validation

Python · LangGraph · Scikit-learn · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_raq

- Achieved **93.6% precision and 87.9% F1** across 10 classification rules on 1,240 documents by building an end-to-end ML pipeline with feature extraction, rule-based classification, and a deterministic validation engine — every prediction is fully traceable.
- Implemented a **pass@k evaluation harness** (pass@1 81.7%, pass@3 88.3%) with multi-trial sampling, regression gate, and scoring across accuracy and reasoning quality — the same model validation rigor required in production financial services ML.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML pipelines, anomaly detection, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

ML & Statistical Modeling: Python, R, Scikit-learn, PyTorch, Random Forest, XGBoost, Logistic Regression, Isolation Forest, Linear & Ridge Regression, Feature Engineering, Cross-Validation, Hypothesis Testing, Statistical Analysis, Model Evaluation, Anomaly Detection, Pandas, NumPy

Data & Cloud: AWS (S3, EC2, RDS, Glue), SQL, PL/SQL, PostgreSQL, MySQL, Spark, Docker, CI/CD (GitHub Actions), ETL Pipelines, Data Governance, Git, Linux

Visualization & Communication: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Interactive Dashboards

Additional: FastAPI, REST APIs, JavaScript, TypeScript, RabbitMQ, Agile/Scrum (120+ sprints), PHI/HIPAA Compliance